In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE(Main) - 2017

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	B	C
2.	D	B	B
3.	B	D	B
4.	D	C	B
5.	B	D	A
6.	B	D	D
7.	A	C	C
8.	A	D	D
9.	A	A	A
10.	D	A	A
11.	D	A	B
12.	C	B	C
13.	B	B	B
14.	C	C	D
15.	D	A	B
16.	C	D	C
17.	A	C	A
18.	B	A	A
19.	A	D	C
20.	C	D	B
21.	A	A	A
22.	D	B	C
23.	D	D	B
24.	D	A	A
25.	A	B	D
26.	C	D	B
27.	B	B	D
28.	B	C	C
29.	A	C	A
30.	D	A	B

Physics

PART – I

SECTION – A

3. $m = 2.5\,kg, F = 80\times10^3\,N$

$\therefore a \Rightarrow \dfrac{F}{m} = 32\times10^3\,m/s^2, v = 0, s = 18\times10^{-3}\,m$

Applying, $v^2 - u^2 = 2as$

$u^2 = 2\times32\times10^3\times18\times10^{-3}$

$u = 24\sqrt{2} \approx 34\,m/s$

6. When the ball is pressed by 'x'

$\Delta V = -ax$, for small 'x' final volume $\approx V$

$\Delta P = -B\dfrac{\Delta V}{V} \Rightarrow \dfrac{Bax}{V}$

$F = -\Delta P\times a \Rightarrow \dfrac{d^2x}{dt^2} = -\dfrac{Ba^2}{Vm}x$

$\therefore \omega = \sqrt{\dfrac{Ba^2}{Vm}}$, $Hence\, T = 2\pi\sqrt{\dfrac{Vm}{Ba^2}}$

8. $N = mg - \dfrac{mV^2}{R}, if\ car\ leaves\ the\ road\ \ N = 0$

$\therefore \dfrac{V^2}{R} = g \quad or\ \ v = \sqrt{gR} = 20\,m/s$

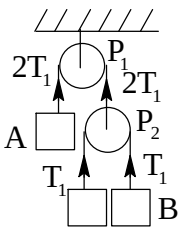
11. $F_{max} - mg = m\omega^2 A \Rightarrow F_{max} = m\left(g + \omega^2 A\right)$

$\omega = 2\pi\times10 \Rightarrow 20\pi\ rod/s, A = \dfrac{80\times10^{-3}}{2} \Rightarrow 40\times10^{-3}\,m$

$m = 400\times10^{-3}\,kg$

Substituting the values we get $F_{max} \Rightarrow 68N$

12.



$5m$

Since 5m is stationary, T_1=5mg

If the acceleration of the pulley $P_2 = a_1 \uparrow$

$a_b = 2a_1 \uparrow (w.r.t.\ ground)$

$\therefore 5mg - mg = 2ma_1 ---------(1)$

If $a_{A/P_1} = a_2,\ a_A = (a_2 - a_1)\ w.r.t.\ ground$

$\therefore 10mg - mg = m(a_2 - a_1) -----(2)$

From equation (1) $a_1 \Rightarrow 2g$

Thus $a_2 = 11g \Rightarrow 110\, m/s^2$

13. Applying Bernoulli's theorem

$$4\times10^5 + P_{atm} = P_{atm} + \frac{1}{2}\rho v^2$$

$$v \Rightarrow \sqrt{\frac{8\times10^5}{10^3}} \Rightarrow 20\sqrt{2} \approx 28\, m/s$$

16. From graphs, $|\in| = \left|-A\times\frac{dB}{dt}\right| \Rightarrow 8\times10^{-4}\times3\times10^{-3}$

$\Rightarrow 24\times10^{-7} V$

Also, $I = 2mA,\ R = \frac{\in}{I} \Rightarrow \frac{24\times10^{-7}}{2\times10^{-3}} \approx 1.2\, m\Omega$

17. $PV^N = k$

$\therefore V^N dP + NPV^{N-1}dV = 0$

Or $VdP = -NP\ dV$

$\therefore \frac{dP}{dV} = \frac{-NP}{V}$, also at $P = 2\times10^5\ P_a\ \&V = 4\times10^5\ m^3$

$\frac{dP}{dV} = \frac{-3}{4} \Rightarrow \therefore N = \frac{4}{2}\times\frac{3}{4} \Rightarrow 1.5$

21. Applying Moseley's Law

$$\upsilon_2 = \upsilon_1\left(\frac{z_2 - 1}{z_1 - 1}\right)^2$$

24. $dU = \frac{1}{2}\frac{B^2}{\mu}dV$

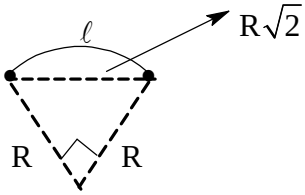
$\Rightarrow \frac{1}{2}\frac{B^2}{\mu}\times 2\pi dr$

$B = \frac{\mu r^2}{a^2}\frac{I}{2\pi r} \Rightarrow \frac{\mu Ir}{2\pi a^2}$

$$dU = \frac{1}{2} \times \frac{1}{\mu} \times \frac{\mu^x I^2 r^2}{4\pi^2 a^4} \times 2\pi r dr$$

$$U \Rightarrow \frac{\mu I^2}{4\pi a^4} \int_0^a r^3 dr \Rightarrow \frac{\mu I^2}{16\pi}$$

29.



Let the pole strength be 'm'

$\therefore M = m\ell$

$$M^1 = mR\sqrt{2},\ R = \frac{2\ell}{\pi}$$

$$\therefore M^1 = \frac{2ml\sqrt{2}}{\pi} \Rightarrow \frac{2\sqrt{2}M}{\pi}$$

Chemistry

PART – II

SECTION – A

2. $I_2 + dil.OH^- \longrightarrow I^- + IO^- + H_2O$

4. $Mg_2C_3 + H_2O \longrightarrow Mg(OH)_2 + C_3H_4$

5. Inert pair effect

6. $a \equiv bonding \qquad C \equiv anti\,bording$
   $b \equiv bonding \qquad D \equiv non-bonding$

8. $T_c = \dfrac{89}{27Rb}$ comes out to be more then $200°C$ option D is correct.

10. $\dfrac{1}{2}H_2(g) + \dfrac{1}{2}Cl_2(g) \longrightarrow HCl(g)$ $\Delta H_f = B.E.of\ reactions - B.E.of\ products$

$= \dfrac{1}{2}(104 + 56) - 103 = -22\,Kcal/mol$

12. $C_xH_y + \left(\dfrac{2x + \frac{4}{2}}{2}\right)O_2 \longrightarrow xCO_2 + \dfrac{4}{2}H_2O$

Volume of $CO_2 = 100 - 60 = 40\,ml$

molar ratio of $CO_2 : H_2O = 40 : 60 = 2 : 3$

13. $BO_3^{3-} + CH_3OH \xrightarrow{H^+} B(OMe)_3$

15. $\underset{(X)}{SO_3^{2-}} + 2H^+ \longrightarrow H_2O + \underset{(Y)}{SO_2}$

SO_2 is a colourless gas with irritating smell.

$3SO_2 + K_2Cr_2O_7 + H_2SO_4 \longrightarrow \underset{(Green)}{Cr_2(SO_4)_3} + K_2SO_4 + H_2O$

17. Because Cr(+VI) has d° configuration hence d-d transition will not possible.

18. d^7 → 1 1 eg

→ ↑↓ ↑↓ ↑ t_2g $t2g^5eg^2$

Total paired electron = 2
Total pairing energy = 2P
CFSE $= -0.8\Delta_0$ (or -8 Dq) $+ 2P$

21. $\Delta T_f = i\,k_f m$

$i = \dfrac{\Delta T_f}{k_f m} = \dfrac{.0054}{1.86 \times .001} = 2.9 \cong 3$

Hence, no. of ions =3

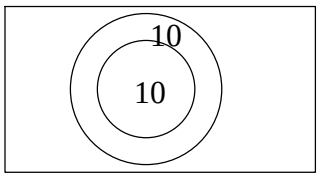
$[\,Pt(NH_3)_4Cl_2] \rightarrow [Pt(NH_3)_4Cl_2]^{2+} + 2Cl^-$

Mathematics

PART – III

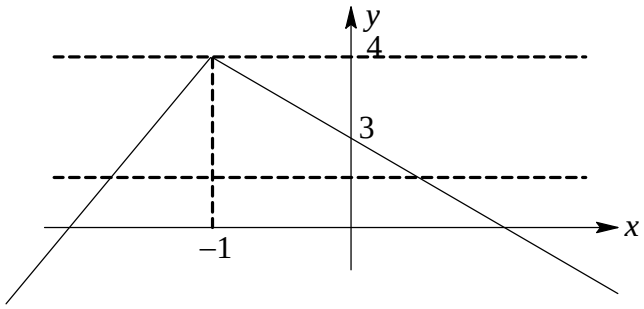
SECTION – A

1. $n(A)=10, n(B)=20$



$n(A\Delta B)=10$

2.



$2^{f(x)}=2^2 \Rightarrow f(x)=2$

4. $Tn=\dfrac{2}{n(n+1)(n+2)}$

$Tn=\dfrac{1}{n(n+1)}-\dfrac{1}{(n+1)(n+2)}$

On solving

$=\dfrac{n(n+3)}{2(n+1)(n+2)} \Rightarrow K=2$

5. $\dfrac{18!}{6!6!6!}$

7. $f(x)=\begin{cases}0 & x\in\mathbb{Z}\\ 1 & x\phi\mathbb{Z}\end{cases}$

$gof=\sin^{-1}0=0$

$\sin^{-1}1=\dfrac{\pi}{2}$

9. $f(x)=[x]+2 \Rightarrow f(g(x))=[\cos x]+2$

$f\left(g\left(\dfrac{\pi}{2}\right)\right)=0$

13. $xy=1$

$$x\frac{dy}{dx}+y=0$$

$$\frac{dy}{dx}=-\frac{y}{x}$$

$$y-y_1=\frac{x_1}{y_1}(x-x_1)$$

$$yy_1-y_1^2=x_1x-x_1^2$$

$$x_1x-yy_1+y_1^2-x_1^2=0$$

14. $$=\begin{vmatrix}1 & p & q\\ 0 & q-p & r-q\\ 0 & r-q & p-r\end{vmatrix}$$

$$=1\left[(q-p)(p-r)-(r-q)(r-q)\right]$$

$$=\frac{-1}{2}\left[(p-q)^2+(q-r)^2+(r-p)^2\right]=0$$

$$\Rightarrow p=q=r$$

$$\angle P=\angle Q=\angle R$$

15. $(t+1)x+8y-4t=0$

$$tx+(t+3)y-(3t-1)=0$$

$$\frac{t+1}{t}=\frac{8}{t+3}\pm\frac{-4t}{-(3t-1)}$$

$$t=3$$

16. $$\begin{vmatrix}4 & -a & -3\\ 1 & 2 & a\\ a & 0 & 2\end{vmatrix}=0$$

$$4(4-0)+a(2-a^2)+3(0-2a)=0$$

$$a^3+4a-16=0$$

$$a=2$$

17. $$I\int\frac{3x^2+2x}{(x^3+x^2+1)^2+2^2}dx$$

$$=\frac{1}{2}\tan^{-1}\left(\frac{x^3+x^2+1}{2}\right)+C$$

18. $f(x)=\int_{1/x}^{\sqrt{x}}\sin\left(t^2\right)dt$

$f'(x)=\sin x.\frac{1}{2\sqrt{x}}-\sin\left(\frac{1}{x^2}\right)\left(\frac{-1}{x^2}\right)$

$f'(1)=\frac{\sin 1}{2}+\sin 1$

$=\frac{3}{2}\sin 1$

19. $y=\left[\frac{x^2}{36}+1\right]$

Area $=\frac{1}{2}(2+3)\times 1$

$=\frac{5}{2}$

20. $(xy-1)\frac{dy}{dx}=-y^2$

$\frac{xy-1}{y^2}=\frac{-dx}{dy}$

$\frac{dx}{dy}=\frac{1}{y^2}-\frac{x}{y}$

$\frac{dx}{dy}+\frac{1}{y}x=\frac{1}{y^2}$

$I.F.=e^{\int\frac{1}{y}dy}=y$

$x.y=\int\frac{1}{y^2}.ydy$

$xy=\ln y$

$ke=\ln e$

$k=\frac{1}{e}$

22. $n(E)={}^3C_2+{}^3C_2+{}^4C_2+{}^3C_1\times{}^4C_1=24$

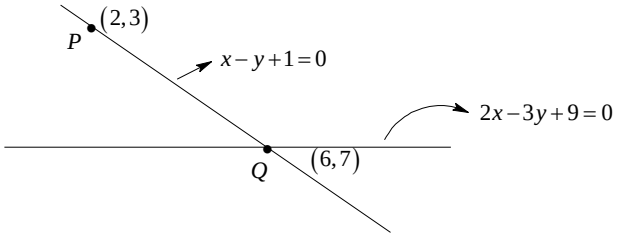
$n(S)={}^{10}C_2=45$

$P(E)=\frac{24}{45}\Rightarrow\frac{8}{15}$

23. $(1+x)^{13}(x+1)^{14}=\left\{{}^{13}C_0+{}^{13}C_1x+{}^{13}C_2x^2+....+{}^{13}C_{13}x^{13}\right\}$

Coeff of x^{13} in $(1+x)^{27}={}^{27}C_{13}={}^{27}C_{14}$

24.



$PQ = \sqrt{16+16}$

$= 4\sqrt{2}$

25. For two common tangents.

$|r_1 - r_2| < C_1C_2 < r_1 + r_2$

$r_1 = 1$

$r_2 = \sqrt{16+9-x^2}$

$C_1C_2 = \sqrt{16+9} = 5$

$\left|\sqrt{25-\lambda^2} - 1\right| < 5 < 6\sqrt{25-\lambda^2} + 1$

$-3 < \lambda < 3$

26. Slope of $PQ = \dfrac{\frac{C}{t_1} - \frac{C}{t_2}}{Ct_1 - Ct_2}$

Slope of $PQ = \dfrac{-1}{t_1t_2}$

Slope of $RS = \dfrac{-1}{t_3t_4}$

$t_1t_2t_3t_4 = -1$

Slope of $CP = \dfrac{\frac{C}{4} - 0}{Ct_1 - 0} = \dfrac{1}{t_1^2}$

$\dfrac{1}{t_1^2} \cdot \dfrac{1}{t_2^2} \cdot \dfrac{1}{t_3^2} \cdot \dfrac{1}{t_4^2} = 1$

27. $\dfrac{z_1 - z_2}{z_3 - z_2} = \left|\dfrac{z_1 - z_2}{z_3 - z_2}\right| e^{i\pi/4}$

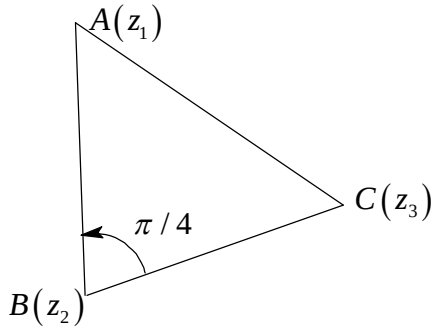
$\dfrac{z_1 - z_2}{z_3 - z_2} = \dfrac{1}{2}(1+i)$

$\Rightarrow 2z_1 - 2z_2 = (1+i)(z_3 - z_2)$

$\Rightarrow 2z_1 - 2z_2 = z_3 - z_2 + i(z_3 - z_2)$

$i(z_2 - z_3) = z_3 - z_2 - 2z_1 + 2z_2$

$= z_3 + z_2 - 2z_1$



29. $CV = \frac{\sigma}{x} \Rightarrow V_1 = \frac{\sigma}{30} \quad V_2 = \frac{\sigma}{25}$

$6V_1 - 5V_2 = \frac{\sigma}{5} - \frac{\sigma}{5} = 0$

30. $\frac{a+b+c}{3} = A \qquad (abc)^{1/3} = G$

$a+b+c = 3A \quad abc = G^3$

$H = \frac{3}{\frac{1}{b}+\frac{1}{b}+\frac{1}{c}} \Rightarrow H = \frac{3abc}{ab+bc+ca} \Rightarrow H = \frac{3G^3}{ab+bc+ca}$

$ab+bc+ca = \frac{3G^3}{H}$

$x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$